

HEARTBLEED

Project Research Paper and Whitepaper

Automated Identity Correlation and Digital Intelligence

Developed by: Alpha-07 (Metis Labs)

June 2026

Part I: Research Paper

Multi-Platform Identity Correlation in Modern OSINT

Abstract

Modern digital life is characterized by fragmentation. Individuals spread their online presence across numerous platforms, often using varying handles, bios, and identifiers. This fragmentation creates a challenge for investigators, security researchers, and OSINT analysts. This paper presents HeartBleed, an automated toolkit designed to solve the identity correlation problem using a modular collection framework and a multi-factor weighted scoring algorithm. We explore the methodology of connecting disparate data points into a single cohesive persona through algorithmic analysis and graph-based visual intelligence.

1. Introduction

In the current era of ubiquitous social media, the 'Digital Footprint' of an individual is rarely contained within a single platform. For an effective OSINT investigation, a researcher must move from a single identifier (like a username) to a full map of the target's presence. Traditional manual methods are time-consuming and prone to human oversight. Automated tools exist, but frequently lack a robust way to evaluate the probability that two discovered profiles belong to the same person. HeartBleed addresses this gap by introducing an objective scoring engine designed specifically for identity correlation.

2. Methodology

2.1 Modular Collection

HeartBleed utilizes a 'Plugin-based' architecture. Each social platform (GitHub, X, Instagram, etc.) is managed by an independent collector module. This decoupling allows the engine to query disparate APIs and public web pages concurrently, normalizing the returned data (Bios, Locations, Websites, Avatars) into a standardized internal model. This modularity ensures the tool remains extensible as new platforms emerge.

2.2 Identity Correlation Algorithm

The core of the research is the Correlation Engine. It evaluates every profile against the original search identifier and against other discovered profiles in the same investigation. Factors included in the weighted score are:

- Exact Username Match (+30 points)
- Similar Username Pattern (+20 points)
- Shared Location Data (+15 points)
- Identical Personal Website (+15 points)
- Perceptual Bio Similarity (+20 points)

By totaling these factors, HeartBleed assigns a Confidence Level (Low to Very High), allowing analysts to focus their

efforts on high-probability leads.

2.3 Visual Intelligence and Graph Theory

Data is most effective when it can be visualized. HeartBleed maps the discovered identity web using directed graphs. In this model, the Target is a central node, and each profile is a peripheral node. Edges (connections) represent the strength of the correlation score. Crucially, the tool identifies 'Hub Nodes' - profiles that are shared across multiple targets - exposing relationships that might be invisible in traditional list-based reports.

3. Conclusion

The results of our tests demonstrate that automated correlation significantly reduces the investigative workload. By automating the 'low-level' collection and providing a 'high-level' probabilistic analysis, HeartBleed allows researchers to move from data collection to active intelligence analysis in minutes rather than hours. The inclusion of multi-target workspaces further enhances the tool's utility in complex, real-world scenarios.

Part II: Whitepaper

HeartBleed: The Next Generation of OSINT Intelligence

Vision and Mission

The mission of the HeartBleed project is to empower the OSINT community with professional-grade, modular, and high-performance tools that operate entirely on public data. We believe in Efficiency, Transparency, and Ethics. Our vision is to build the world's most accessible and extensible identity correlation framework for the security industry.

Core Architecture: Clean and Resilient

HeartBleed is built with Python 3.11+ using 'Clean Architecture' principles. This ensures the tool is lightweight and capable of running on everything from high-powered Linux servers to mobile Android environments (Termux). The codebase is divided into:

- Core Orchestrator: Manages concurrency and work-flow.
- Collectors: Independent scrapers and API clients.
- Analyzers: Local intelligence processing engines.
- Reporters: Multi-format export system (HTML, JSON, PDF).

Feature Roadmap

v0.1: Core Engine, Concurrent Collectors, and HTML Graph Visualization.

v0.2: Username Mutation Engine, local Bio Profiler, and Digital Dorker.

v0.3: Email Discovery, Domain Intelligence, and PDF Exporting.

v1.0 (Future): Automated image reverse-search, Dark Web integration, and AI-Persona Summarization.

Ethical Framework

HeartBleed is an OSINT (Open Source Intelligence) tool. It is strictly governed by the principle of Public Access. The toolkit DOES NOT bypass authentication, harvest private credentials, or exploit software vulnerabilities. All information collected originates from publicly accessible APIs, meta tags, and web pages. Users of HeartBleed are expected to adhere to legal requirements and respect the terms of service of the platforms they analyze.

Developer Ecosystem

The project is designed for community growth. Adding a new platform to HeartBleed is a simple 3-step process: (1) Create a new class in the collectors folder, (2) Implement the 'fetch' method, and (3) Define the normalization logic. This low barrier to entry allows the tool to stay ahead of the changing social media landscape.